JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY

Mysuru – 570 006.

SRS AND DESIGN DOCUMENT ON:

"JOBGUARD AI"

AI-Assisted Job Scam Detection and Risk Assessment System

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY:-

NAME	USN
Charan M	01JST24UBC024
Chethan Kumar L E	01JST24UBC025

UNDER THE SUPERVISION OF:-

Dr. C. S. Santhosh

Assistant Professor

Department of Computer Application

2026–2027

DEPARTMENT OF COMPUTER APPLICATIONS

1. DATASET AND PREPROCESSING

1.1 Study Area

The study is based on cybersecurity, mainly the detection of fake or fraudulent job opportunities found through online recruitment. A scam can appear in the job description, recruiter communication, application link or payment request. Hence, the study considers job text, recruiter details, URLs, company information and other details normally available during an application.

JobGuard AI is developed as a full-stack Progressive Web Application. The user provides the available job details and the system sends the required information through different analysis stages. This document mainly describes the data and model pipeline used to identify suspicious jobs.

1.2 Dataset Sample and Dataset Description

Two public datasets are used for the machine-learning part of JobGuard AI. EMSCAD contains 17,880 job advertisements arranged across 18 source columns, including the target label. The job-related text and fraud label support text classification; other metadata columns are not included in the TF-IDF vocabulary. PhiUSIIL contains 235,795 URL records and 54 documented features. The implemented URL model retains 18 features that can be calculated from the URL string; the other 36 columns are not used at runtime. User-entered job details and links are analysed at runtime and are not added to these benchmark training records. [11, 12]

Dataset / Source	Purpose	Size / Composition	JobGuard AI Use
EMSCAD fake job postings	Text scam classification	17,880 rows × 18 columns	Selected job text + fraud label → TF-IDF + Logistic Regression
PhiUSIIL phishing URL dataset	URL phishing classification	235,795 rows × 54 features	18 URL-string features retained; 36 features not used by runtime URL model

1.2.1 Representative Sample Dataset

The screenshots below show the source dataset columns and sample records. The representative table illustrates the kinds of inputs used by JobGuard AI; the rows are examples to explain the input format and are not additional benchmark records.

Sample ID	Input Type	Representative Content / Attribute	Label / Use
S01	Job text	Entry-level role with unusually high pay and an advance registration request	Fraud-indicator example
Sample ID	Input Type	Representative Content / Attribute	Label / Use

S02	Job text	Conventional role description with normal hiring information and no fee request	Legitimate-pattern example
S03	URL	Application link containing suspicious lexical/structural characteristics	URL-risk example
S04	URL	Secure careers URL with normal domain structure	Lower-risk URL example
S05	Live input	Company + recruiter contact + vacancy + URL	Integrated analysis

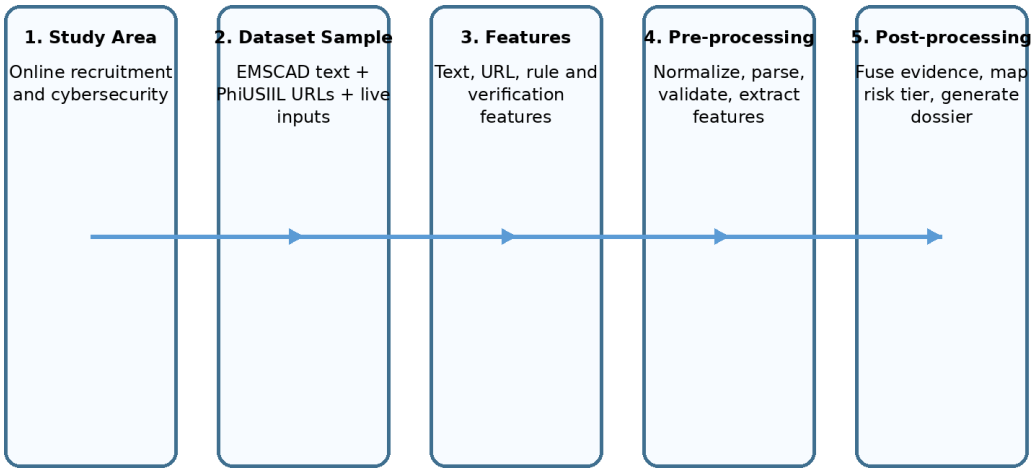
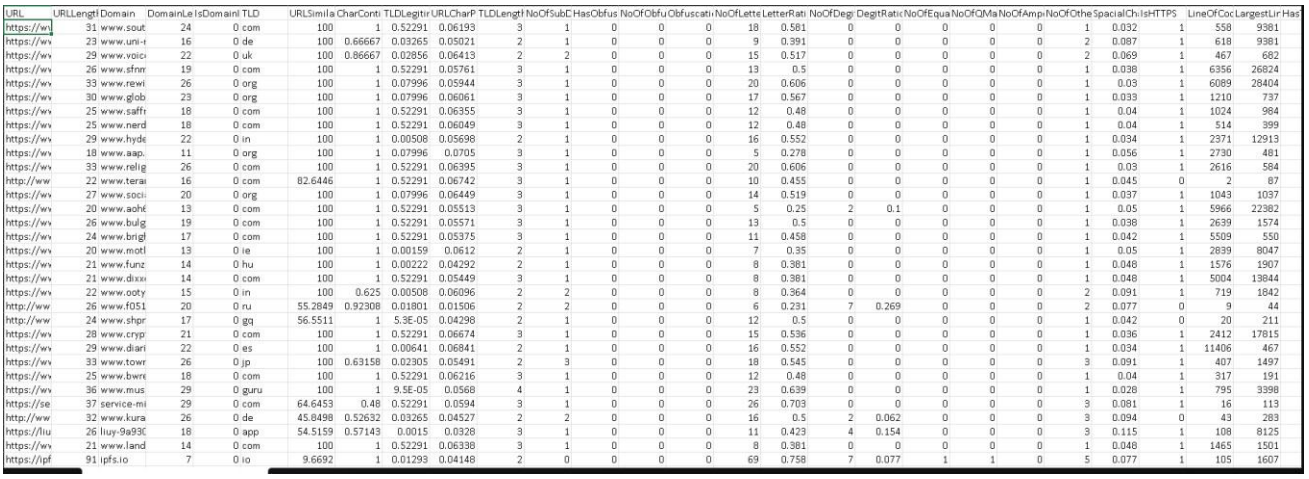


Figure 1.1. Dataset Workflow

job_id	title	location	department	salary_ran	company_	description	requireme	benefits	telecommi	has_comp	has_questi	employme	required_e	required_e	industry	function	fraudulent
1	Marketing	US, NY, Ne	Marketing		We're Foo	Food52, a	Experience with conte		0	1	0	Other	Internship			Marketing	0
2	Customer	NZ, , Auckl	Success		90 Second	Organised	What we e	What you	0	1	0	Full-time	Not Applicable		Marketing	Customer	0
3	Commissic	US, IA, Wever			Valor Servi	Our client,	Implement pre-comm		0	1	0						0
4	Account E:	US, DC, W:	Sales		Our passio	THE COMF	EDUCATIC	Our cultur	0	1	0	Full-time	Mid-Senio	Bachelor's	Computer	Sales	0
5	Bill Review	US, FL, Fort Worth			SpotSourc	JOB TITLE:	QUALIFICA	Full Benefi	0	1	1	Full-time	Mid-Senio	Bachelor's	Hospital &	Health Car	0
6	Accounting	US, MD,				Job Overview	Apex is an environr		0	0	0						0
7	Head of C	DE, BE, Ber	ANDROID	20000-280	Founded ir	Your Resp	Your Know	Your Bene	0	1	1	Full-time	Mid-Senio	Master's D	Online Me	Manageme	0
8	Lead Gues	US, CA, San Francisco			Airenvyâ€'	Who is Air	Experience	Competitiv	0	1	1						0
9	HP BSM SN	US, FL, Pensacola			Solutions3	Implement	MUST BE A US CITIZE		0	1	1	Full-time	Associate		Information	Technolo	0
10	Customer	US, AZ, Phoenix			Novitex Er	The Custor	Minimum Requiremer		0	1	0	Part-time	Entry level	High Schoc	Financial S	Customer	0
11	ASP.net De	US, NJ, Jersey City		100000-120000		Position : I	Position : I	Benefits -	0	0	0	Full-time	Mid-Senio	Bachelor's	Informatic	Informatic	0
12	Talent Sou	GB, LND, L	HR			Want to bi	TransferW	Weâ€™re	You will jo	0	1	0					0
13	Applicatior	US, CT, Stamford			Novitex Er	The Applic	Requirements:x4 â€“ 5		0	1	0	Full-time	Associate	Bachelor's	Manageme	Informatic	0
14	Installers	US, FL, Orlando			Growing e	Event Indu	Valid driver's license,S		0	1	1	Full-time	Not Applic	Unspecifie	Events Ser	Other	0
15	Account E:	AU, NSW, '	Sales		Adthena is	Are you int	Youâ€™ll	In return w	0	1	0	Full-time	Associate	Bachelor's	Internet	Sales	0
16	VP of Sale:	SG, 01, Sin	Sales	120000-15	Jungle Ven	About Vau	Key Superp	Basic: SGD	0	1	1	Full-time	Executive	Bachelor's	Facilities S	Sales	0
17	Hands-On	IL, , Tel Avi	R&D		At HoneyB	We are loc	Previous experience i		0	1	0	Full-time	Mid-Senior	level	Internet	Engineerin	0
18	Southend-	GB, SOS, Southend-on-Sea			Establishe	Governme	16-18 year	Career pro	0	1	1						0
19	Visual Desi	US, NY, New York			Kettle is ar	Kettle is hiring a Visual Designer!			0	1	0						0
20	Process Cc	US, PA, USA Northeast			We Provid	Experience	Must have 5 or more		0	0	0	Full-time					0
21	Marketing	US, TX, Austin			IntelliBrigh	IntelliBrigh	Job Requirements	Assi	0	1	0					Marketing	0
22	Front End	NZ, N, Auckland			Frustrated	Want to be	You will m	You will be	0	1	0	Full-time	Mid-Senio	Master's D	Consumer	Engineerin	0
23	Engageme	AE, ,	Engagement		Upstreamâ	The positic	Requireme	Salary &ar	0	1	1	Full-time	Mid-Senio	Bachelor's	Telecomm	Sales	0
24	Vice Presic	US, CA, Ca	Businessfri	100000-12	WDM Grot	#URL_eda	Job Requir	Businessfri	0	1	0	Full-time	Executive	Unspecifie	Internet	Sales	0
25	Customer	GB, LND, London				We are a canary wharf based e-c			0	0	0						0

Figure 1.2(a). EMSCAD dataset screenshot.



URL	URLLength	Domain	DomainLe	IsDomainI	TLD	URLSimila	CharConti	TLDLegitir	URLCharP	TLDLength	NoOfSubD	HasObfus	NoOfObfu	Obfuscati	NoOfLette	LetterRati	NoOfDegi	DegitRatic	NoOfEqua	NoOfQM	NoOfAmpi	NoOfOthe	SpacialCh	IsHTTPS	LineOfCoc	LargestLir	Has
https://w	31	www.sout	24	0	com	100	1	0.52291	0.06193	3	1	0	0	0	18	0.581	0	0	0	0	0	1	0.032	1	558	9381	
https://w	23	www.uni-t	16	0	de	100	0.66667	0.03265	0.05021	2	1	0	0	0	9	0.391	0	0	0	0	0	2	0.087	1	618	9381	
https://w	29	www.voic	22	0	uk	100	0.86667	0.02856	0.06413	2	2	0	0	0	15	0.517	0	0	0	0	0	2	0.069	1	467	682	
https://w	26	www.sfnm	19	0	com	100	1	0.52291	0.05761	3	1	0	0	0	13	0.5	0	0	0	0	0	1	0.038	1	6356	26824	
https://w	33	www.rewi	26	0	org	100	1	0.07996	0.05944	3	1	0	0	0	20	0.606	0	0	0	0	0	1	0.03	1	6089	28404	
https://w	30	www.glob	23	0	org	100	1	0.07996	0.06061	3	1	0	0	0	17	0.567	0	0	0	0	0	1	0.033	1	1210	737	
https://w	25	www.saffr	18	0	com	100	1	0.52291	0.06355	3	1	0	0	0	12	0.48	0	0	0	0	0	1	0.04	1	1024	984	
https://w	25	www.nerd	18	0	com	100	1	0.52291	0.06049	3	1	0	0	0	12	0.48	0	0	0	0	0	1	0.04	1	514	399	
https://w	29	www.hyde	22	0	in	100	1	0.00508	0.05698	2	1	0	0	0	16	0.552	0	0	0	0	0	1	0.034	1	2371	12913	
https://w	18	www.aap.	11	0	org	100	1	0.07996	0.0705	3	1	0	0	0	5	0.278	0	0	0	0	0	1	0.056	1	2730	481	
https://w	33	www.relig	26	0	com	100	1	0.52291	0.06395	3	1	0	0	0	20	0.606	0	0	0	0	0	1	0.03	1	2616	584	
http://ww	22	www.terai	16	0	com	82.6446	1	0.52291	0.06742	3	1	0	0	0	10	0.455	0	0	0	0	0	1	0.045	0	2	87	
https://w	27	www.soci	20	0	org	100	1	0.07996	0.06449	3	1	0	0	0	14	0.519	0	0	0	0	0	1	0.037	1	1043	1037	
https://w	20	www.aohé	13	0	com	100	1	0.52291	0.05513	3	1	0	0	0	5	0.25	2	0.1	0	0	0	1	0.05	1	5966	22382	
https://w	26	www.bulg	19	0	com	100	1	0.52291	0.05571	3	1	0	0	0	13	0.5	0	0	0	0	0	1	0.038	1	2639	1574	
https://w	24	www.brigl	17	0	com	100	1	0.52291	0.05375	3	1	0	0	0	11	0.458	0	0	0	0	0	1	0.042	1	5509	550	
https://w	20	www.motl	13	0	ie	100	1	0.00159	0.0612	2	1	0	0	0	7	0.35	0	0	0	0	0	1	0.05	1	2839	8047	
https://w	21	www.funz	14	0	hu	100	1	0.00222	0.04292	2	1	0	0	0	8	0.381	0	0	0	0	0	1	0.048	1	1576	1907	
https://w	21	www.dixx	14	0	com	100	1	0.52291	0.05449	3	1	0	0	0	8	0.381	0	0	0	0	0	1	0.048	1	5004	13844	
https://w	22	www.ooty	15	0	in	100	0.625	0.00508	0.06096	2	2	0	0	0	8	0.364	0	0	0	0	0	2	0.091	1	719	1842	
http://ww	26	www.f051	20	0	ru	55.2849	0.92308	0.01801	0.01506	2	2	0	0	0	6	0.231	7	0.269	0	0	0	2	0.077	0	9	44	
http://ww	24	www.shpr	17	0	gq	56.5511	1	5.3E-05	0.04298	2	1	0	0	0	12	0.5	0	0	0	0	0	1	0.042	0	20	211	
https://w	28	www.cryp	21	0	com	100	1	0.52291	0.06674	3	1	0	0	0	15	0.536	0	0	0	0	0	1	0.036	1	2412	17815	
https://w	29	www.diari	22	0	es	100	1	0.00641	0.06841	2	1	0	0	0	16	0.552	0	0	0	0	0	1	0.034	1	11406	467	
https://w	33	www.towr	26	0	jp	100	0.63158	0.02305	0.05491	2	3	0	0	0	18	0.545	0	0	0	0	0	3	0.091	1	407	1497	
https://w	25	www.bwre	18	0	com	100	1	0.52291	0.06216	3	1	0	0	0	12	0.48	0	0	0	0	0	1	0.04	1	317	191	
https://w	36	www.mus	29	0	guru	100	1	9.5E-05	0.0568	4	1	0	0	0	23	0.639	0	0	0	0	0	1	0.028	1	795	3398	
https://se	37	service-mi	29	0	com	64.6453	0.48	0.52291	0.0594	3	1	0	0	0	26	0.703	0	0	0	0	0	3	0.081	1	16	113	
http://ww	32	www.kura	26	0	de	45.8498	0.52632	0.03265	0.04527	2	2	0	0	0	16	0.5	2	0.062	0	0	0	3	0.094	0	43	283	
https://liu	26	liuy-9a93(	18	0	app	54.5159	0.57143	0.0015	0.0328	3	1	0	0	0	11	0.423	4	0.154	0	0	0	3	0.115	1	108	8125	
https://w	21	www.land	14	0	com	100	1	0.52291	0.06338	3	1	0	0	0	8	0.381	0	0	0	0	0	1	0.048	1	1465	1501	
https://ipf	91	ipfs.io	7	0	io	9.6692	1	0.01293	0.04148	2	0	0	0	0	69	0.758	7	0.077	1	1	0	5	0.077	1	105	1607	

Figure 1.2(b). PhiUSIIL dataset screenshot.

1.3 Features

JobGuard AI uses separate feature groups because job descriptions and URLs contain different signals. Text features represent useful words and phrases in a posting; URL features describe the address structure; rule features record suspicious requests; and verification features check whether company, vacancy and contact details agree with available public sources.

Feature Group	Technique / Source	Important Features
Text features	TF-IDF n-gram representation	Term frequency, inverse document frequency, vocabulary terms, n-grams
URL features	Lexical and structural URL parsing	Length, domain length, IP flag, TLD length, subdomains, obfuscation, letters, digits, special characters, HTTPS

1.3.1 Text Feature Extraction

EMSCAD contains 17,880 rows and 18 source columns, including the fraud label. Before extraction, the selected model inputs are the job-description text and its label; the other source columns are not part of the TF-IDF vocabulary. After text feature extraction, the row count remains 17,880 and the configured TF-IDF vector width is 12,000 features. The model is evaluated with 14,304 training records and 3,576 held-out testing records. The report does not include the saved fitted vectorizer vocabulary, so 12,000 is the configured vector width rather than a separately measured active-term count. These numeric text features are used by Logistic Regression to estimate whether a posting is fraudulent. [11]

1.3.2 URL Feature Extraction

PhiUSIIL contains 235,795 rows and 54 documented source features. After URL feature extraction, JobGuard AI retains 18 numeric features calculated directly from each URL; 36 other source columns are excluded from the runtime URL model. The resulting URL feature matrix therefore contains 235,795 rows × 18 model-input columns before the train/test split. An 80:20 stratified split produces 188,636 training records and 47,159 test records. The 18 values describe URL length, domain structure, obfuscation, character patterns and HTTPS use, and are supplied to HistGradientBoostingClassifier. [12, 15]

Dataset	Before extraction	Technique	After extraction	Used for
EMSCAD	17,880 rows × 18 source columns	Select job text; TF-IDF word n-grams	17,880 rows × 12,000 configured TF-IDF dimensions; label kept separately	Logistic Regression
PhiUSIIL	235,795 rows × 54 source features	Parse URL and calculate lexical/structural values	235,795 rows × 18 selected URL features; 36 source features excluded	HistGradientBoostingClassifier

Count interpretation: feature extraction changes the column representation, not the number of benchmark records. The EMSCAD text pipeline is configured for a 12,000-dimensional TF-IDF vector; the exact active vocabulary count can only be confirmed from the fitted vectorizer artifact. PhiUSIIL uses 18 selected URL input features from the 54 documented source features.

Table 1.4. Feature extraction and reduction used in JobGuard AI

1.4 Preprocessing and Post-processing

1.4.1 Pre-processing Steps

The project documentation does not specify any fixed number of rows deleted during cleaning, so no deletion count is claimed. The benchmark row counts are retained for reporting: 17,880 EMSCAD job advertisements and 235,795 PhiUSIIL URL records before the stated train/test splits. Preprocessing cleans and separates the source information, then produces model-ready text, metadata and URL features.

1. Clean extra spaces and normalize the text.
2. Identify URLs, e-mail addresses, phone numbers, salary values and messaging handles.
3. Separate job-description text from company and contact information.
4. Convert the cleaned text into TF-IDF n-gram features.
5. Parse the application URL and calculate the 18 selected structural features.
6. Keep unsupported webpage or HTML features outside the runtime URL model.
7. Run the fixed scam and payment rules along with the statistical models.
8. Perform public-source verification only when the required security checks allow it.

Pre-processing output: normalized job text, extracted contact/payment entities, structured metadata and a validated URL represented by 18 numeric URL features.

1.4.2 Feature Extraction Techniques Used

Technique	Purpose	Stage
Pattern / regular-expression extraction	Identify URLs, emails, phone numbers, salary expressions and messaging handles	Pre-processing
TF-IDF n-gram extraction	Convert normalized recruitment text into numerical features	Feature extraction
URL lexical/structural parsing	Generate the 18 URL-only model features	Feature extraction
Rule-based keyword/pattern matching	Generate interpretable scam and payment evidence	Feature extraction
OCR / text extraction when applicable	Extract text from recruitment screenshots or documents	Input processing

Technique	Purpose	Stage
Public-source matching	Verify company, vacancy and recruitment lifecycle evidence	Verification

1.4.3 Post-processing Steps

1. Collect the outputs from the text, URL, rule and verification modules.
2. Keep each component result separately so the final decision can be explained.
3. Combine the available evidence using the selected risk-fusion weights.
4. Convert the combined value into a 0–100 risk score and threat level.
5. Add verification states such as VERIFIED, NOT FOUND, MISMATCH or UNABLE TO VERIFY.
6. Prepare the final result with the score, reasons, verification information and safety advice.

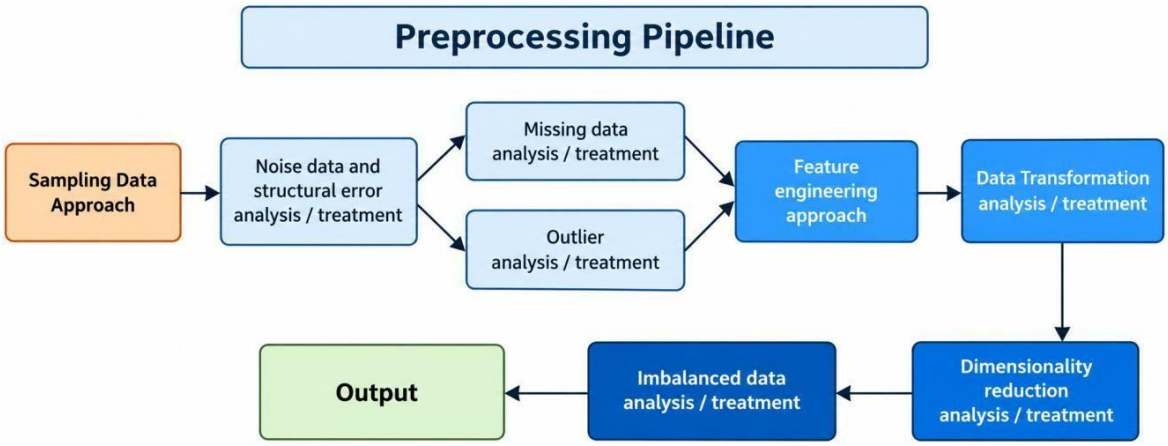


Figure 1.3. Data Preprocessing Pipeline

Models and Algorithms

The models and algorithms below describe how JobGuard AI converts job text and application URLs into model evidence, and how deterministic rules contribute interpretable scam indicators. Each algorithm is followed by its equation and implementation steps.

2.1 Rule-Based Scam Detection Algorithm

The rule engine checks warning signs that can be identified directly from the job information. Examples include registration fees, security deposits, cheque overpayment, cryptocurrency or wallet payments, movement to Telegram or WhatsApp, unusually high salary claims and pressure to act quickly. Its main benefit is that the triggered reason can be shown clearly to the user. [11]

$$\text{RuleScore} = \Sigma_i \, (w_i \times I_i)$$

Equation 2.1. Deterministic Rule-Score Formulation

Where: w_i = rule weight; I_i = rule indicator (1 if triggered, otherwise 0); RuleScore = total rule evidence; Σ_i = sum over all rules.

Algorithm Steps

1. Receive normalized text and extracted structured metadata.
2. Check each configured recruitment-scam and payment rule.
3. Set the corresponding indicator to 1 when the rule condition is satisfied.
4. Multiply each triggered indicator by its configured severity weight.
5. Aggregate the rule contributions and preserve the matched evidence.
6. Pass the structured rule evidence to the risk-fusion stage.

Use in JobGuard AI

JobGuard AI uses a deterministic rule engine to examine indicators such as advance fees, suspicious payment methods, cheque-overpayment schemes, messaging-only communication, unrealistic salaries, and urgent hiring pressure.

The main advantage is explainability: when a rule is triggered, the system can identify the corresponding warning sign.

Unlike a machine learning classifier, the rule engine does not need to learn every condition from labelled training data. However, it can miss unfamiliar scams that do not match its predefined rules.

Difference from research-based classification: The rule engine complements machine learning by identifying known warning signs directly. It is not a replacement for the TF-IDF text classifier or URL classification model.

2.2 TF-IDF Text Representation

TF-IDF stands for Term Frequency–Inverse Document Frequency. It converts the words in a job advertisement into numbers that can be used by the classifier. Words that help distinguish suspicious and normal postings receive suitable weights. In this project, sub-linear term frequency is used with a maximum vocabulary of 12,000 features. [13]

$$\mathrm{TF}(t,d) = 1 + \log(\mathrm{count}(t,d))$$

Equation 2.2. Sub-linear Term-Frequency

Where: t = term/token; d = document/job posting; count(t,d) = occurrences of t in d; TF(t,d) = sub-linear term frequency.

$$\mathrm{IDF}(t) = \log(N \,/\, (1 + \mathrm{df}(t)))$$

Equation 2.3. Inverse Document Frequency

Where: N = total training documents; df(t) = documents containing t; IDF(t) = inverse document-frequency weight.

$$\text{TF-IDF}(t,d) = \mathrm{TF}(t,d) \times \mathrm{IDF}(t)$$

Equation 2.4. TF-IDF Feature Weight

Where: TF-IDF(t,d) = final term weight; TF(t,d) = term-frequency part; IDF(t) = inverse document-frequency part.

Algorithm Steps

1. Normalize the recruitment text.
2. Tokenize the text and generate the configured n-grams.
3. Calculate sub-linear term-frequency values.
4. Calculate inverse document-frequency values from the training corpus.
5. Multiply TF and IDF values to form the sparse TF-IDF vector.
6. Send the vector to the trained Logistic Regression classifier.

Use in JobGuard AI

JobGuard AI uses TF-IDF to convert cleaned job-related text into a numerical representation that the Logistic Regression classifier can process.

The documented pipeline configures the vectorizer for a maximum vocabulary size of 12,000 features. This is the **configured maximum feature dimension**, not proof that all 12,000 features occur in the fitted vocabulary.

The process is:

1. Receive the job-related text.
2. Clean and normalize the text.
3. Tokenize the text into terms or n-grams.
4. Calculate TF-IDF weights.
5. Generate a numerical feature vector.
6. Pass the vector to the Logistic Regression classifier.

Difference from previous research

TF-IDF is an established text-representation technique rather than a new algorithm developed for this project. Its role in JobGuard AI is to represent job advertisements numerically so that the classifier can identify patterns associated with legitimate and fraudulent postings.

The project uses TF-IDF as one stage of a larger system that also examines URL features and predefined scam indicators.

2.3 Logistic Regression Classifier

Logistic Regression is applied to the TF-IDF values. It combines the feature values using learned weights and uses the sigmoid function to obtain a probability. This probability becomes one of the evidence values used in the final JobGuard AI decision. [14]

$$z = \beta_0 + \Sigma_i \beta_i x_i$$

Equation 2.5. Logistic Linear Score

Where: β_0 = intercept; β_i = learned weight for feature i; x_i = feature value; z = linear score.

$$P(y=1|x) = 1 / (1 + e^{-z})$$

Equation 2.6. Logistic Sigmoid Probability

Where: P(y=1|x) = positive-class probability; e = Euler's constant; z = linear score; y=1 = positive class.

Algorithm Steps

1. Load the trained TF-IDF vocabulary and Logistic Regression coefficients.
2. Transform the normalized job text using the same TF-IDF representation used during training.
3. Compute the linear score from the feature vector and learned coefficients.
4. Apply the sigmoid function to obtain the class probability.
5. Retain the probability and relevant model evidence.
6. Pass the text-model result to the consensus risk-fusion layer.

Use in JobGuard AI

JobGuard AI applies Logistic Regression to the TF-IDF representation of job-related text.

The process is:

1. Load the trained TF-IDF vectorizer and Logistic Regression model.
2. Convert the incoming job text into the expected numerical representation.
3. Calculate the weighted linear score.
4. Apply the sigmoid function.
5. obtain the model's class probability or prediction.
6. Pass the text-model evidence to the risk-fusion component.

Difference from previous research

Logistic Regression is an established classification algorithm, not a new algorithm invented for JobGuard AI.

The project's contribution is its use as the text-classification component of a larger job-scam detection pipeline. Its output is considered alongside rule-engine evidence, URL analysis, and other available assessment information.

The final result therefore does not depend exclusively on the Logistic Regression prediction.

2.4 URL Feature Extraction

The URL extractor works directly on the supplied URL. It calculates the selected structural values without downloading the whole webpage. This keeps the URL model simple and makes the runtime analysis less dependent on external page content. [12]

No.	URL Feature	Meaning
1	URLLength	Total length of the URL.
2	DomainLength	Length of the domain component.
3	IsDomainIP	Whether the domain is represented as an IP address.
4	TLDLength	Length of the top-level domain.
5	NoOfSubDomain	Number of subdomain components.

No.	URL Feature	Meaning
6	HasObfuscation	Whether obfuscation is detected.
7	NoOfObfuscatedChar	Count of obfuscated characters.
8	ObfuscationRatio	Ratio of obfuscated characters.
9	NoOfLettersInURL	Number of alphabetic characters.
10	LetterRatioInURL	Ratio of letters in the URL.
11	NoOfDigitsInURL	Number of digits.
12	DigitRatioInURL	Ratio of digits.
13	NoOfEqualsInURL	Count of '=' characters.
14	NoOfQMarkInURL	Count of '?' characters.
15	NoOfAmpersandInURL	Count of '&' characters.
16	NoOfOtherSpecialCharsInURL	Count of other special characters.
17	SpecialCharRatioInURL	Ratio of special characters.
18	IsHTTPS	Whether HTTPS is used.

Algorithm Steps

1. Validate and parse the supplied URL.
2. Separate domain, path, query and other lexical components.
3. Calculate the 18 configured structural features.
4. Validate feature values and maintain the expected feature order.
5. Create the numerical feature vector for the URL model.
6. Pass the vector to HistGradientBoostingClassifier. [15]

Use in JobGuard AI

JobGuard AI extracts 18 URL-related features from the supplied URL:

The process is:

1. Receive the URL.
2. Examine its structure and character patterns.
3. Calculate the selected numerical features.
4. Arrange the features into a model-input vector.
5. Pass the vector to the URL classification model.

This component analyses URL structure rather than downloading and examining the entire webpage. Consequently, its findings should be interpreted as URL-based evidence, not as a complete verification of the website.

Difference from previous research

The PhiUSIIL research uses a broader set of 54 documented features and proposes a security-profile-based approach to phishing URL detection. JobGuard AI retains 18 URL-related inputs for its documented runtime URL model.

This is an important distinction: **JobGuard AI does not use all 54 PhiUSIIL features in its runtime URL model.** It uses a selected subset focused on URL structure and character patterns.

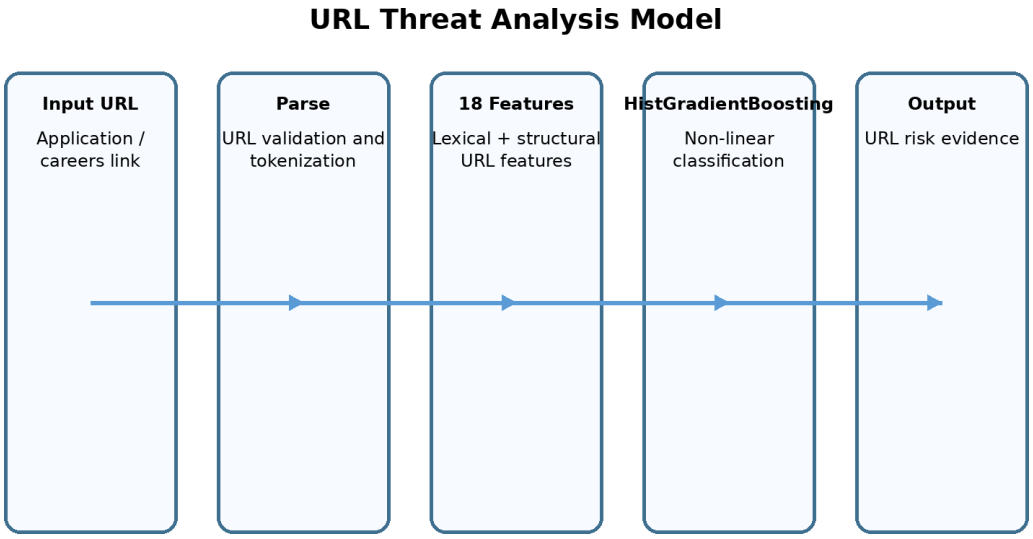


Figure 2.2. URL Threat Analysis Model

2.7 Algorithm Summary

Algorithm / Engine	Input	Main Output	Role
Rule Engine	Normalized text + metadata	Deterministic evidence	Explainable scam/payment indicators
TF-IDF	Normalized job text	Sparse numerical vector	Text representation
Logistic Regression	TF-IDF vector	Class probability	Text scam evidence
URL Feature Extractor	Application URL	18-feature vector	URL structural representation

2.8 Research Background and Algorithm Selection

Research on online recruitment fraud has explored machine learning techniques for identifying fraudulent job advertisements. Researchers have used datasets containing legitimate and fraudulent job postings to study the characteristics that distinguish genuine vacancies from scams.

One important dataset is the Employment Scam Aegean Dataset (EMSCAD), introduced by Vidros and colleagues. It contains 17,880 job advertisements, including legitimate and fraudulent postings. Researchers have used this dataset to investigate the characteristics of recruitment fraud and develop automated detection methods.

Another relevant dataset is the PhiUSIIL Phishing URL Dataset, introduced by Prasad and Chandra. It contains 235,795 URL records represented by 54 documented features. The associated research investigates

phishing URL detection using a security profile that includes URL and webpage characteristics, similarity-based detection, and incremental learning.

JobGuard AI uses these research datasets as the basis for different parts of its detection system. EMSCAD supports the job-text classification component, while PhiUSIIL provides data for the URL classification component. The project combines these machine learning components with a deterministic rule engine, AI-generated explanations, and a risk-fusion mechanism.

The purpose is not to claim that every algorithm used in JobGuard AI is new. Instead, the project integrates established techniques into a single application so that users can examine suspicious job descriptions and URLs and receive a consolidated risk assessment.

Why these algorithms were selected

The algorithms and techniques were selected for the following reasons:

1. **Rule-based detection:** Identifies known scam indicators and provides understandable reasons for warnings.
2. **TF-IDF:** Converts textual job information into numerical features suitable for machine learning.
3. **Logistic Regression:** Classifies job-related text using the numerical features generated by TF-IDF.
4. **URL feature extraction:** Identifies suspicious characteristics such as unusual URL length, obfuscation, domain structure, and HTTPS usage.
5. **HistGradientBoosting:** Analyses the extracted URL features to support URL threat classification.
6. **Risk fusion:** Combines different sources of evidence into one risk assessment.
7. **Gemini AI:** Generates a structured explanation of the assessment, including warning signs and recommended actions.

These techniques perform different functions. Their combination allows JobGuard AI to examine several types of evidence rather than depending entirely on one classification model.

DESIGN

This design section explains the system workflow, processing stages, overall architecture, risk fusion and performance evaluation. Separate text, URL, rule and verification results are combined to produce a risk score and an explanation of the decision.

3. PROPOSED METHODOLOGY AND PERFORMANCE METRICS

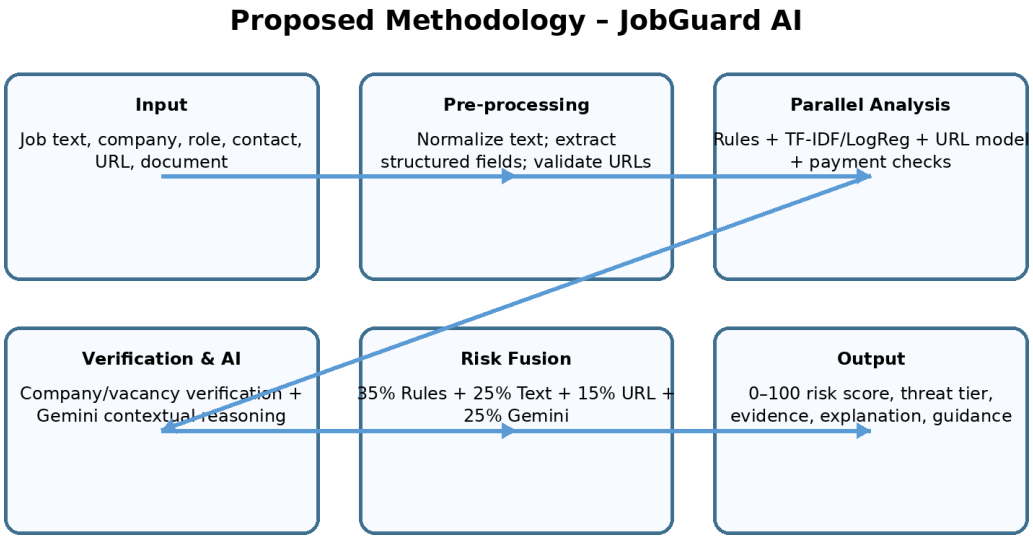


Figure 3.1. Proposed Methodology – JobGuard AI

3.1 Block Diagram and Workflow

The complete workflow is shown in the block diagram below. The user input is first cleaned and separated into the parts needed by the different analysis modules. Their results are then checked together before the final result is displayed.

The block diagram begins with heterogeneous recruitment input. Pre-processing prepares the data for independent analytical engines. Rule-based detection, text classification, URL analysis and payment checks operate as complementary evidence streams. Public verification and Gemini contextual reasoning enrich the evidence before risk fusion.

3.1.1 Methodology Steps

1. Collect the job description, company name, role, recruiter details, application URL and available documents.
2. Clean the text and identify URLs, e-mail addresses, phone numbers and payment-related expressions.
3. Convert the job text into TF-IDF features and classify it using Logistic Regression.
4. Parse the application URL and calculate the 18 selected URL features before URL classification.
5. Check the job information using the fixed scam and payment rules.
6. Compare company, vacancy, domain and recruiter information with available public sources.
7. When network access is available, use contextual AI reasoning to interpret the collected evidence.
8. Combine the available evidence using the selected risk-fusion weights.
9. Display the risk score, threat level, main reasons, verification status and safety guidance.

3.2 Input, Intermediate Processing and Output

Stage	Input	Middle Process	Output
1. Input ingestion	Job text, company, role, contact, URL, document	Validate fields and identify input types	Structured input package
2. Pre-processing	Raw text / links	Normalize, parse, extract entities	Clean text + metadata
3. Text analysis	Clean job text	TF-IDF → Logistic Regression	Text probability/evidence
4. URL analysis	Application URL	18-feature extraction → HistGradientBoosting	URL risk evidence
5. Rule/payment analysis	Text + metadata	Pattern matching and severity scoring	Deterministic evidence
6. Verification	Company, vacancy, domain/contact	Safe public-source checks	Verification states
7. Contextual reasoning	Structured evidence	Gemini interpretation when connected	Contextual evidence
8. Risk fusion	Available evidence streams	Weighted consensus + safeguards	0–100 risk score + threat tier
9. Final output	Fused evidence	Format explanation and guidance	Structured threat dossier

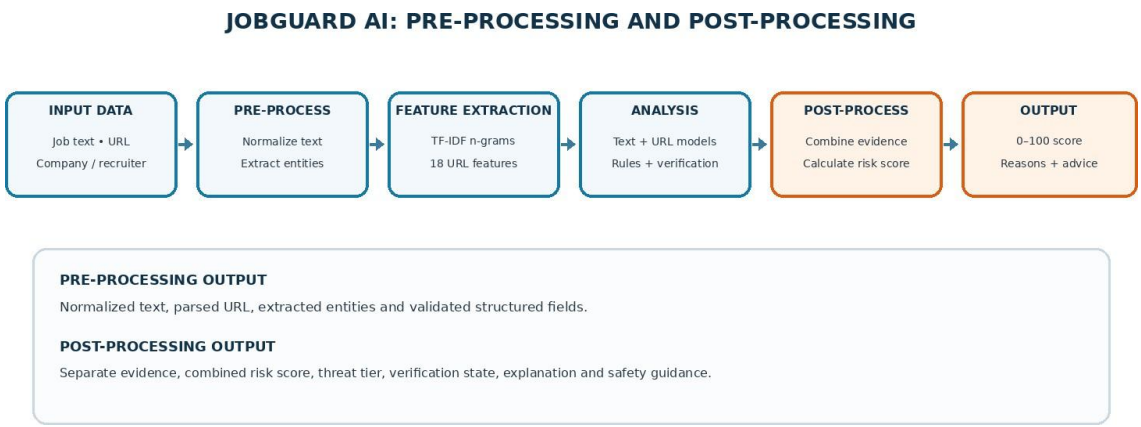


Figure 3.3. Pre-processing and Post-processing Workflow

3.2.1 Risk-Fusion Formulation

$$R = 100 \times (0.35H + 0.25M + 0.15U + 0.25G)$$

Equation 3.1. Baseline JobGuard AI Risk-Fusion Formulation

Where: R = final 0–100 risk score; H = rule/payment evidence; M = text-model evidence; U = URL-model evidence; G = Gemini evidence. The four weights are fixed engineering weights.

3.3 Model Training and Testing

For the text model, EMSCAD is cleaned first and then converted into TF-IDF features before Logistic Regression is trained. The source dataset contains 17,880 job advertisements and 18 columns. The vectorizer is configured for 12,000 TF-IDF dimensions; the fitted active vocabulary count is not separately recorded in this document. An 80:20 stratified split gives 14,304 training records and 3,576 held-out testing records. [11]

For the URL model, PhiUSIIL has 235,795 records and 54 documented features. After extraction, 18 URL-only features are retained and 36 other source columns are not used by the runtime URL model. An 80:20 stratified split gives 188,636 training records and 47,159 testing records. [12]

Table 3.3. Dataset Collection and Feature-Selection Summary

Dataset	Collected records	Original fields / features	Retained for JobGuard AI	Excluded / not used
EMSCAD	17,880 rows	18 source columns	Job text transformed to a configured 12,000-dimensional TF-IDF vector; fraud label retained separately	Non-text metadata excluded from TF-IDF vocabulary
EMSCAD split	14,304 train / 3,576 test	Same feature schema in both subsets	14,304 training rows and 3,576 held-out rows; each vector uses the same fitted vocabulary	Test rows excluded from model fitting
PhiUSIIL	235,795 rows	54 source features	18 URL-derived model input columns after extraction	36 source features not used by runtime URL model
PhiUSIIL split	188,636 train / 47,159 test	Same 18-feature schema in both subsets	188,636 training rows and 47,159 held-out rows	Test rows excluded from model fitting

3.4 Overall Architecture and Layer Description

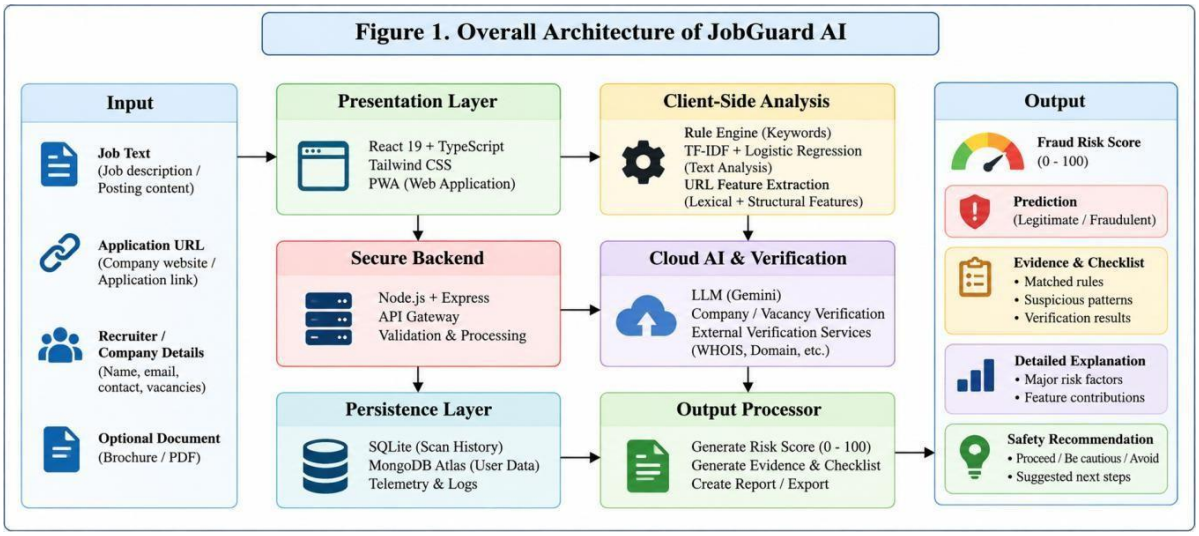


Figure 3.4. Overall Architecture of JobGuard AI

Architecture output: a 0–100 risk score, threat tier, text/URL/rule evidence, company and vacancy verification status, plain-language explanation and recommended safety actions. The output dossier is the final stage of the architecture.

Architecture layer description

1. Input layer: The user/job seeker supplies the job description, application URL, recruiter or company details, and optional supporting documents. Missing fields are handled as unavailable evidence rather than assumed facts.

2. Presentation layer: The React and TypeScript interface presents the forms, scan workflow, progress state and final risk dossier. Tailwind CSS supports responsive layouts, while PWA capabilities provide installable app behavior.

3. Client-side analysis layer: The rule engine checks known scam patterns, TF-IDF and Logistic Regression analyse job text, and URL feature extraction prepares the 18 structural URL values for URL classification. This layer creates early evidence without requiring every step to call a cloud service.

4. Secure backend and credentials/validation layer: The Node.js and Express API receives scan requests, validates inputs and coordinates protected server-side operations. API credentials and external-service secrets should remain on the server and must not be exposed in the browser. The gateway also controls which external checks may be requested.

5. Cloud AI and verification layer: When an internet connection and valid configuration are available, Gemini interprets the structured evidence. Public company and vacancy sources are checked to compare the employer, domain, role and recruitment status. Unavailable checks are marked as unable to verify.

6. Persistence layer: SQLite stores scan history and local records, while MongoDB Atlas can store cloud telemetry or synchronized records when configured. Persistence supports reviewing previous scans; it does not by itself prove that a job is legitimate.

7. Output dossier layer: The final result presents a 0–100 risk score, threat tier, separate text/URL/rule evidence, company and vacancy verification state, a plain-language explanation, and recommended safety actions. The output is an assessment to support user judgment, not a guarantee.

3.5 Performance Metrics

Performance evaluation uses accuracy, precision, recall, F1-score and ROC-AUC. The equations define each measure; actual scores should be reported only after evaluating the held-out test data. [16, 17]

3.5.1 Classification Metrics

3.5.1.1 Confusion Matrix Terms

Term	Meaning
TP (True Positive)	Fraud/phishing sample correctly classified as positive.
TN (True Negative)	Legitimate sample correctly classified as negative.
FP (False Positive)	Legitimate sample incorrectly classified as positive.

Term	Meaning
FN (False Negative)	Fraud/phishing sample incorrectly classified as negative.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Equation 3.2. Accuracy

Where: Accuracy = overall correct classification; TP = true positive; TN = true negative; FP = false positive; FN = false negative.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Equation 3.3. Precision

Where: Precision = correctness of predicted positives; TP = true positives; FP = false positives.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Equation 3.4. Recall / Sensitivity

Where: Recall = correctly detected actual positives; TP = true positives; FN = false negatives.

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Equation 3.5. F1-Score

Where: F1 = harmonic mean of precision and recall; Precision = positive-prediction correctness; Recall = positive-class detection coverage.

ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) summarizes ranking performance across classification thresholds. Higher values indicate stronger separation between the positive and negative classes for the evaluated dataset.

$$\textbf{ROC-AUC} = \int_0^1 \text{TPR}(\text{FPR})\, d(\text{FPR})$$

Equation 3.6. ROC-AUC

Where: TPR = true positive rate (recall); FPR = false positive rate; AUC = area under the ROC curve.

The PhiUSIIL evaluation uses an 80:20 stratified split, producing a held-out test set of 47,159 records. System-level evaluation also considers verification correctness, evidence completeness, security controls, behavior under missing network conditions and usability of the generated risk dossier.

ROC-AUC is evaluated from prediction scores across thresholds. No numeric ROC-AUC value is claimed here because a measured result was not supplied.

REFERENCES

1. Adebowale, M. A., Lwin, K. T., & Hossain, M. A. (2024). Intelligent phishing and advance-fee fraud detection in digital hiring using graph neural networks. IEEE Transactions on Neural Networks and Learning Systems, 35(4), 5120–5133.

2. Al-Ahmadi, S., Al-Bander, B., & Al-Nuaimi, A. (2020). Detecting fake job postings on social media using machine learning models. Journal of Big Data, 7(1), 1–19.

3. Google Cloud. (2025). Gemini API documentation and structured reasoning architecture reference. Google AI for Developers.

4. Habib, A., Al-Khateeb, H., & Epiphaniou, G. (2022). Detection of fake job offers on LinkedIn and professional platforms via multi-feature classification. Computers & Security, 114, 102601.

5. Khatri, S., Kaushik, A., & Singhal, A. (2022). Supervised machine learning algorithms for employment scam identification on digital job boards. International Journal of Information Security, 21(3), 645–661.

6. Kumar, P., & Sharma, R. (2025). Hybrid multi-modal framework for real-time recruitment fraud detection with explainable AI. IEEE Transactions on Dependable and Secure Computing, 22(1), 312–326.

7. Lal, S., Tiwari, P., & Garg, S. (2021). Online recruitment fraud detection using deep learning and bidirectional LSTM networks. Expert Systems with Applications, 178, 115012.

8. MongoDB Inc. (2025). MongoDB Atlas multi-cloud database architecture reference manual. MongoDB Documentation.

9. Nagaraj, S., Ramesh, V., & Senthil, K. (2021). Comparative analysis of deep learning classifiers for online job scam detection. Neural Computing and Applications, 33(18), 12045–12059.

10. Raza, M., Qayyum, A., & Ahmad, N. (2023). Transformer-based fake job detection using RoBERTa and contextualized attention mechanisms. Information Processing & Management, 60(3), 103289.

11. Vidros, S., Kolias, C., Kambourakis, G., & Akoglu, L. (2017). Automatic Detection of Online Recruitment Frauds: Characteristics, Methods, and a Public Dataset. Future Internet, 9(1), 6. https://doi.org/10.3390/fi9010006.

12. Prasad, A., & Chandra, S. (2024). PhiUSIIL Phishing URL (Website) [Dataset]. UCI Machine Learning Repository. https://doi.org/10.1016/j.cose.2023.103545.

13. Manning, C. D., Raghavan, P., & Schütze, H. (2008). Introduction to Information Retrieval. Cambridge University Press. (TF-IDF and term weighting).

14. Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). Applied Logistic Regression (3rd ed.). Wiley. (Logistic regression).

15. Friedman, J. H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. The Annals of Statistics, 29(5), 1189–1232. https://doi.org/10.1214/aos/1013203451.

16. Fawcett, T. (2006). An introduction to ROC analysis. Pattern Recognition Letters, 27(8), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010.

17. scikit-learn developers. (n.d.). Model evaluation: precision, recall, F-measure and ROC AUC. https://scikit-learn.org/stable/modules/model_evaluation.html.

18. Vidros, S., Kolias, C., Kambourakis, G., and Akoglu, L. (2017). "Automatic Detection of Online Recruitment Frauds: Characteristics, Methods, and a Public Dataset." *Future Internet*, 9(1), Article 6. EMSCAD research paper

19. Prasad, A., and Chandra, S. (2024). "PhiUSIIL: A diverse security profile empowered phishing URL detection framework based on similarity index and incremental learning." *Computers & Security*, 136, 103545. PhiUSIIL research paper

20. UCI Machine Learning Repository. "PhiUSIIL Phishing URL (Website) Dataset." PhiUSIIL dataset documentation

TABLE OF CONTENTS

Section	Title	Page
1	Software Requirements Specification	2
1.1	Study Area	2
1.2	Dataset Sample and Dataset Description	2
1.2.1	Representative Sample Dataset	2
1.3	Features and Feature Extraction	4
1.3.1	Text Feature Extraction	4
1.3.2	URL Feature Extraction	4
1.4	Preprocessing and Post-processing	5
1.4.1	Pre-processing Steps	5
1.4.2	Feature Extraction Techniques Used	6
1.4.3	Post-processing Steps	6
2	Models and Algorithms	7
2.1	Rule-Based Scam Detection	7
2.2	TF-IDF Text Representation	7
2.3	Logistic Regression Classifier	8
2.4	URL Feature Extraction	9
3	Design	11
3.1	Proposed Methodology and Workflow	11
3.2	Input, Processing and Output	12
3.2.1	Risk-Fusion Formulation	12
3.3	Model Training and Testing	13
3.4	Overall Architecture and Layer Description	14
3.5	Performance Metrics	15
	References	17

LIST OF TABLES

Table	Title	Page
Table 1.1	Dataset Summary	2
Table 1.2	Representative Sample Dataset Format	3
Table 1.3	Feature Groups	4
Table 1.4	Feature Extraction and Reduction	5
Table 2.1	Algorithm Summary	10
Table 3.1	End-to-End Input–Process–Output	12
Table 3.3	Dataset Collection and Feature Selection	13

LIST OF FIGURES

Figure	Title	Page
Figure 1.1	Dataset Workflow	3
Figure 1.2(a)	EMSCAD Dataset Screenshot	3
Figure 1.2(b)	PhiUSIIL Dataset Screenshot	4
Figure 1.3	Data Preprocessing Pipeline	6
Figure 1.4	Pre-processing and Post-processing Workflow	7
Figure 2.2	URL Threat Analysis Model	10
Figure 3.1	Proposed Methodology – JobGuard AI	11
Figure 3.2	Overall Architecture of JobGuard AI	14